

§14.8 Lagrange Multipliers

Homework : 1,3,7,11,15,19,23,39

Problem : Maximize or Minimize functions $f(x, y)$ or $f(x, y, z)$ under certain constraints.

cases	1	2	3
objective function	$f(x, y)$	$f(x, y, z)$	$f(x, y, z)$
constraints	$g(x, y) = k$	$g(x, y, z) = k$	$g(x, y, z) = k$ and $h(x, y, z) = k$

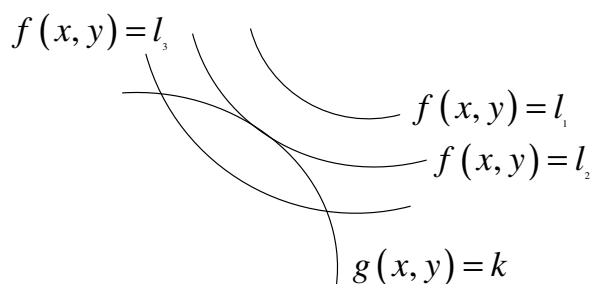
Lagrange Multipliers : 解上述問題的一個方法

想法 :

(i) 極值產生的地方 :

f (變動) 的 level curve or level surface 和
 g (固定) 的 level curve or level surface 相切.

(ii) Level curve of $f \perp \nabla f$.



由 (i) + (ii) $\Rightarrow \nabla f \parallel \nabla g$

結論 : Extreme occur when

(i) $\nabla f = \lambda \nabla g$ for some λ (cases 1 and 2).

(ii) $\nabla f = \lambda \nabla g + \mu \nabla h$ for some λ, μ (case 3).

($\because \nabla f$ 躺在由 ∇g 和 ∇h 所張出來的平面上.)

Example 1 : A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such box.

Solution :

(Method I)

$$f(x, y, z) = xyz$$

$$\text{constraint } g : g(x, y, z) = 2xy + 2yz + xz \quad (1)$$

$$\nabla f = \langle yz, xz, xy \rangle$$

$$\nabla g = \langle 2y + z, 2x + 2z, 2y + x \rangle$$

$$\nabla f // \nabla g \Rightarrow \begin{cases} yz = \lambda(2y + z) \\ xz = \lambda(2x + 2z) \\ xy = \lambda(2y + x) \end{cases}$$

$$\Rightarrow \begin{cases} \frac{y}{x} = \frac{2y + z}{2x + 2z} \\ \frac{z}{y} = \frac{2x + 2z}{2y + x} \\ \frac{x}{z} = \frac{2y + x}{2y + z} \end{cases} \Rightarrow \begin{cases} 2y = x & (2) \\ z = 2y & (3) \\ x = z \end{cases}$$

將(2),(3)代入(1)

$$\Rightarrow 4y^2 + 4y^2 + 4y^2 = 12$$

$$\Rightarrow y = 1$$

$$\Rightarrow x = 2 \quad z = 2$$

$$\Rightarrow V = 4$$

(Method II)

$$\frac{12}{3} = \frac{2xy + 2yz + xz}{3} \geq (4x^2 y^2 z^2)^{\frac{1}{3}}$$

$$\Rightarrow 64 \geq 4x^2 y^2 z^2$$

$$\Rightarrow 4 \geq xyz$$

$$\Rightarrow \text{Maximum volume} = 4.$$

Example2 : $f(x, y) = x^2 + 2y^2$. $g(x, y) = x^2 + y^2 = 1$.

(Method I)

$$\text{高中解法} \begin{cases} 1. \text{算幾不等式} \\ 2. \text{柯西不等式} \\ 3. \text{配方法} \end{cases}$$

$$\Rightarrow f(x, y) = x^2 + 2y^2 = 1 + y^2, \quad (-1 \leq y \leq 1).$$

$$\Rightarrow f \text{ 的最大值} = 2 \quad ; \quad f \text{ 的最小值} = 1.$$

(Method II)

$$\nabla f = \langle 2x, 4y \rangle, \quad \nabla g = \langle 2x, 2y \rangle$$

$$\Rightarrow \begin{cases} x = \lambda x \\ 2y = \lambda y \end{cases} \Rightarrow x = 0 \text{ or } y = 0.$$

$$\Rightarrow (0, \pm 1) \text{ and } (\pm 1, 0)$$

$$\Rightarrow \max f = 2 \quad \min f = 1.$$

Example 3 : $f(x, y, z) = x + 2y + 3z$. Constraints : $\begin{cases} x - y + z = 1 & (1) \\ x^2 + y^2 = 1 & (2) \end{cases}$

(Method I)

$$f(x, y, z) \stackrel{(1)}{=} x + 2y + 3(1 - x + y) = -2x + 5y + 3$$

柯西不等式 得：

$$(x^2 + y^2)((-2)^2 + 5^2) \geq (-2x + 5y)^2$$

$$\Rightarrow \sqrt{29} \geq -2x + 5y \geq -\sqrt{29}$$

$$\Rightarrow \sqrt{29} + 3 \geq -2x + 5y + 3 \geq -\sqrt{29} - 3$$

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Maximum

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minimum

(Method II)

$$\begin{aligned} \nabla f &= \langle 1, 2, 3 \rangle \\ \nabla g &= \langle 1, -1, 1 \rangle \\ \nabla h &= \langle 2x, 2y, 0 \rangle \end{aligned} \Rightarrow \begin{cases} 1 = \lambda + 2\mu x \\ 2 = -\lambda + 2\mu y \\ 3 = \lambda \end{cases}$$

$$\Rightarrow \begin{cases} 2\mu x = -2 \\ 2\mu y = 5 \end{cases} \Rightarrow \frac{x}{y} = \frac{-2}{5}$$

$$\text{Since } x^2 + y^2 = 1 \Rightarrow x = \frac{\mp 2}{\sqrt{29}}, \quad y = \frac{\pm 5}{\sqrt{29}}$$

$$\Rightarrow z = 1 \pm \frac{7}{\sqrt{29}}$$

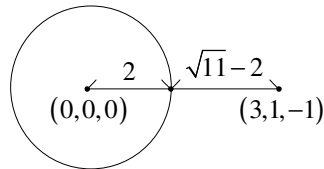
$$\Rightarrow \max = 3 + \sqrt{29}, \quad \min = -3 - \sqrt{29}$$

Example 4 : $x^2 + y^2 + z^2 = 4$. Find max or min of

$$(x-3)^2 + (y-1)^2 + (z+1)^2. \text{ Where those extreme occur?}$$

Solution :

(Method I)



$$\Rightarrow \min = \sqrt{11} - 2 \text{ occurs at } \left(\frac{6}{\sqrt{11}}, \frac{2}{\sqrt{11}}, \frac{-2}{\sqrt{11}} \right) \text{ and}$$

$$\max = \sqrt{11} + 2 \text{ occurs at } \left(\frac{-6}{\sqrt{11}}, \frac{-2}{\sqrt{11}}, \frac{2}{\sqrt{11}} \right).$$

$$\text{Example 5 : } \begin{cases} f(x, y, z) = x + y + z \\ \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1 \end{cases}$$

Solution :

$$\left((\sqrt{x})^2 + (\sqrt{y})^2 + (\sqrt{z})^2 \right) \left(\left(\frac{1}{\sqrt{x}} \right)^2 + \left(\frac{1}{\sqrt{y}} \right)^2 + \left(\frac{1}{\sqrt{z}} \right)^2 \right) \geq (1+1+1)^2 = 9$$

$$\Rightarrow (x + y + z) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) = (x + y + z) \geq 9$$

$$\Rightarrow \text{minimum} = 9.$$

$$\text{Example 6 : } \begin{cases} f(x, y, z) = x^2 + 2y^2 + 2z^2 \\ x + y + z = 1 & (1) \\ x - y + 2z = 2 & (2) \end{cases}$$

Solution :

(Method I)

$$\langle 1, 1, 1 \rangle \times \langle 1, -1, 2 \rangle = \langle 3, -1, -2 \rangle$$

$$\text{Let } z=0 \Rightarrow \begin{cases} x+y=1 \\ x-y=2 \end{cases} \Rightarrow x=\frac{3}{2}, y=-\frac{1}{2}.$$

$\Rightarrow (1) \cap (2)$ 之直線方程式為

$$\begin{cases} x = \frac{3}{2} + 3t \\ y = -\frac{1}{2} - t \\ z = 0 - 2t \end{cases}, \quad t \in \mathbb{R}. \quad (3)$$

將 (3) 代入 f 利用配方法可得極值.

(Method II) Lagrange Multipliers

Example 7 : The plane $x + y + 2z = 2$ intersects the paraboloid $z = x^2 + y^2$ in an ellipse. Find the points on this ellipse that are nearest to and farthest from the origin.

Solution :

$$\text{Max or Min } \begin{cases} f(x, y, z) = x^2 + y^2 + z^2 \\ x + y + 2z = 2 \\ z = x^2 + y^2 \end{cases} \quad \begin{matrix} h \\ \\ g \end{matrix}$$

$$\nabla h = \langle 1, 1, 2 \rangle, \quad \nabla g = \langle 2x, 2y, -1 \rangle, \quad \nabla f = \langle 2x, 2y, 2z \rangle$$

$$\Rightarrow \nabla f = \lambda \langle 1, 1, 2 \rangle + \mu \langle 2x, 2y, -1 \rangle$$

$$\Rightarrow x = y$$

$$\Rightarrow x = \frac{1}{2} \text{ or } -1$$

$$\Rightarrow \text{Two points to check are } \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right) \text{ and } (-1, -1, 2).$$

$$\Rightarrow f\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) = \frac{3}{4} \quad \& \quad f(-1, -1, 2) = 6$$

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最近值

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最遠值